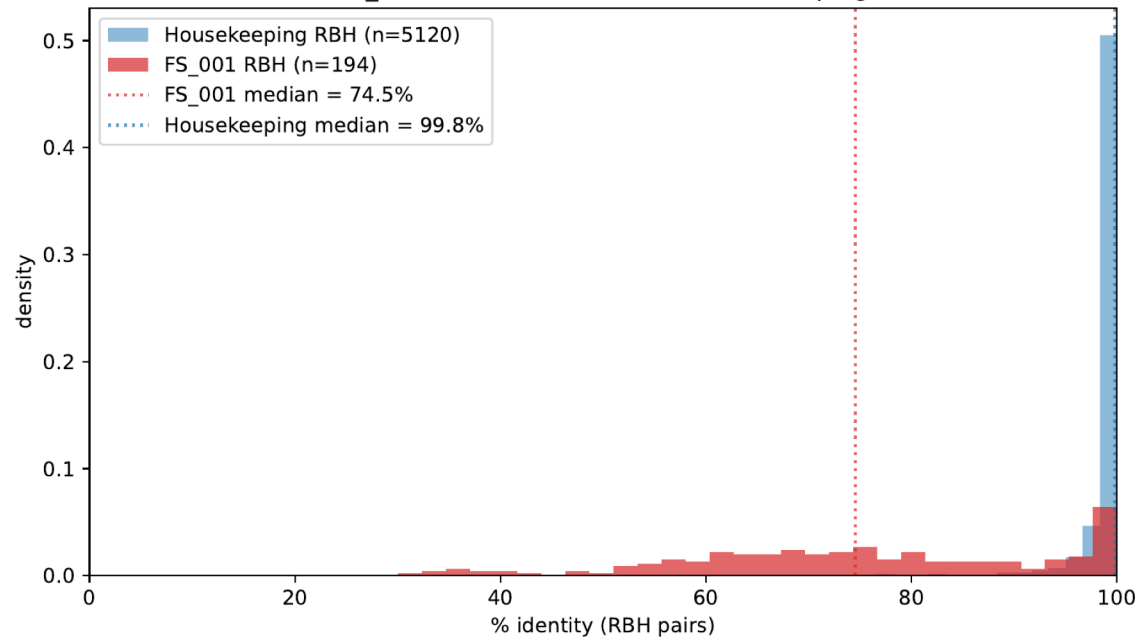


a Sequence Identity in the Twilight Zone

Pairwise amino acid identity of housekeeping core (n=5,120) vs. VIR FS_001 (n=194)

PvW1 ↔ PvPAM ortholog identity

FS_001 (subtelomeric VIR) vs housekeeping



Key Orthology Findings:

- Housekeeping Backbone: Mean identity ~98%, establishing high cross-continental assembly fidelity.
- VIR Antigens in the Twilight Zone: Median identity is only ~32% within FS_001, where sequence alignment methods lose statistical sensitivity.
- Pseudogene Rescue: Translating unannotated GFF3 pseudogenes rescued 145 loci, restoring syntenic landmarks in Peru and Indonesia.
- Syntenic Filtering: Out of 194 reciprocal best-hit (RBH) pairs between Thailand and Peru, only 11 possess conserved flanking microsynteny.

b Cross-Continental Ribbon Synteny Map Between PvW1 (Thailand) and PvPAM (Peru)

Conserved internal chromosomal backbone (grey ribbons) flanked by highly scrambled, non-syntenic subtelomeric VIR clusters

FS_001 ortholog ribbon synteny — PvW1 (top) vs PvPAM (bottom)

All 14 main chromosomes concatenated end-to-end; vertical ticks = FS_001 gene positions; ribbons = RBH ortholog pairs

